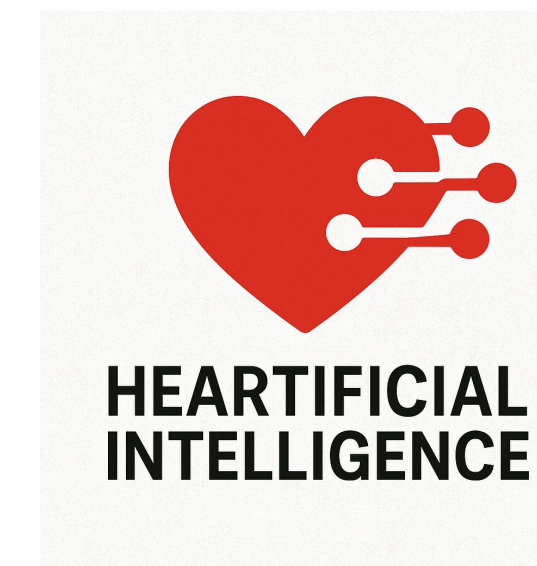




Convolutional Neural Networks for Detection of Chronic Chagas Disease Cardiomyopathies from Electrocardiogram Waveforms

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Background

- Chagas disease affects 7+ million people, primarily in rural subpopulations of Central and South America, and causes cardiomyopathies and complications in a third of patients¹.
- Chagas disease parasitology undetected in the early acute phase often progresses to a chronic phase marked by intracellular amastigotes, arrhythmias, and heart failure²⁻³.
- Chronic Chagas cardiomyopathies are detectable on a 12-lead ECG⁴, and fast deep learning detection could enable targeted confirmatory testing, faster treatment, and vector control.
- Here we preprocess ECG data and develop MLP, CNN, and LSTM models and compare classification performance on The George B. Moody PhysioNet Challenge 2025.

Data

- The George B. Moody PhysioNet Challenge 2025 provides 320,000+ minimally-processed ECG recordings from the CODE-15% dataset, SaMi-Trop dataset, and PTB-XL dataset with sampling frequencies 400-500 Hz⁵.
- Raw ECG data had variant length between 4-10 seconds. For short recordings we padded with 0's on either end of the recording
- We z-score normalized ECG segments to adjust the mean and decrease standard deviation, standardized the time series because of variable sampling Hz

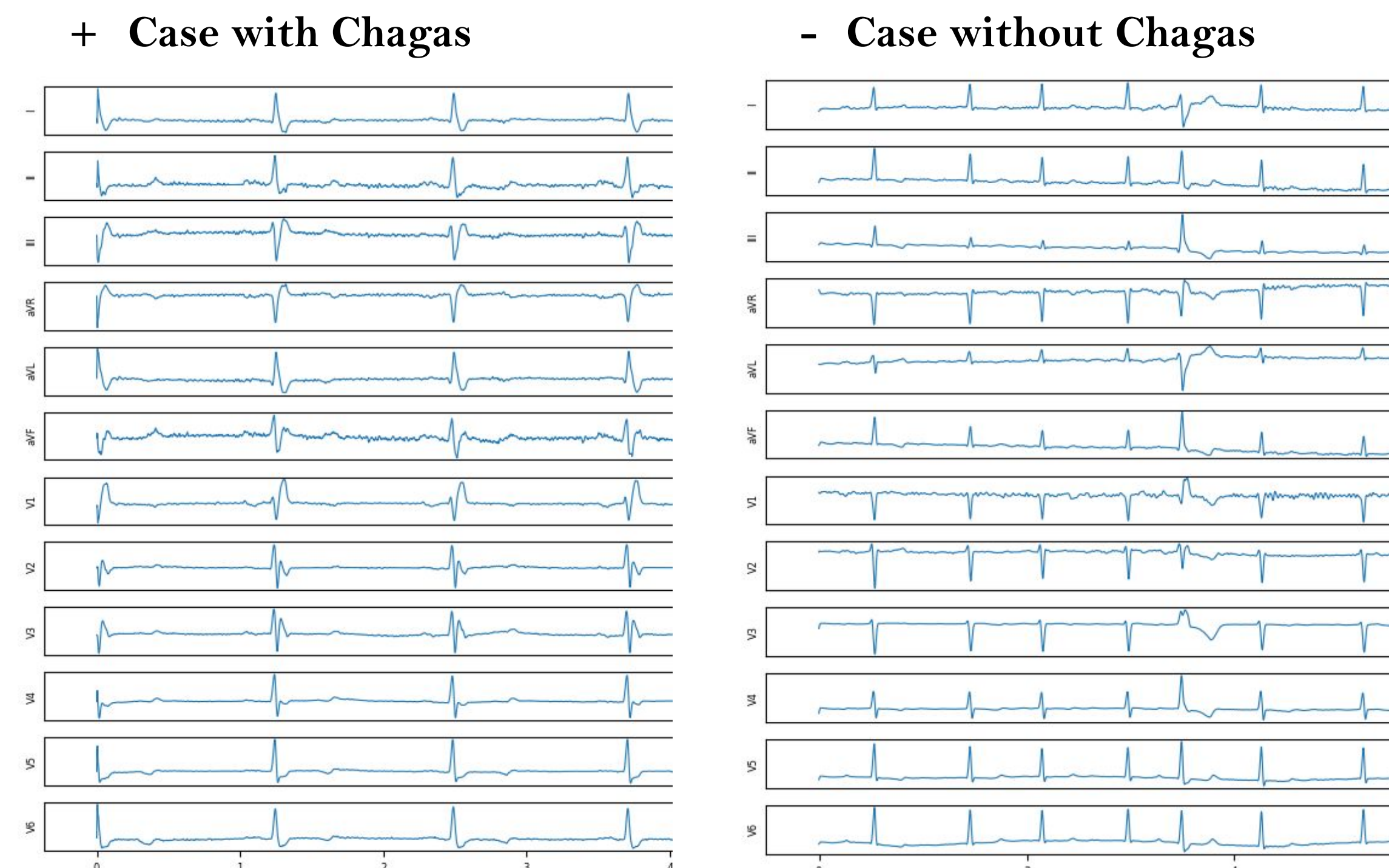


Figure 1. Raw data plotted for 12-lead ECG data from +) Chagas case and -) control participant.

Methodology

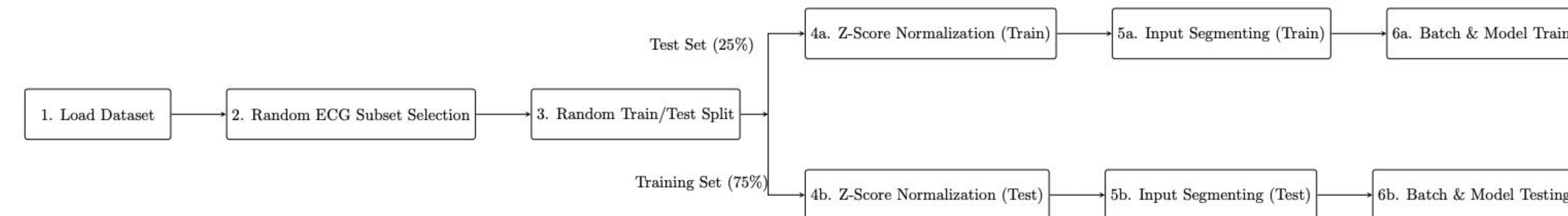


Figure 2: Standardized workflow for training and testing

- Since our 320,000+ samples only contained 6561 true-labeled samples, we randomly sampled 13,122 false-labeled ECG segments to create a ratio of 2:1 negative to positive cases for model training to control the ratio between true false labels.
- We also attempted noising and denoising techniques to improve model with no success
- Sparse convolutional layers aim to compensate for relative uniformity of data between cases and controls
- Experimented with many different architectures but found simple was better

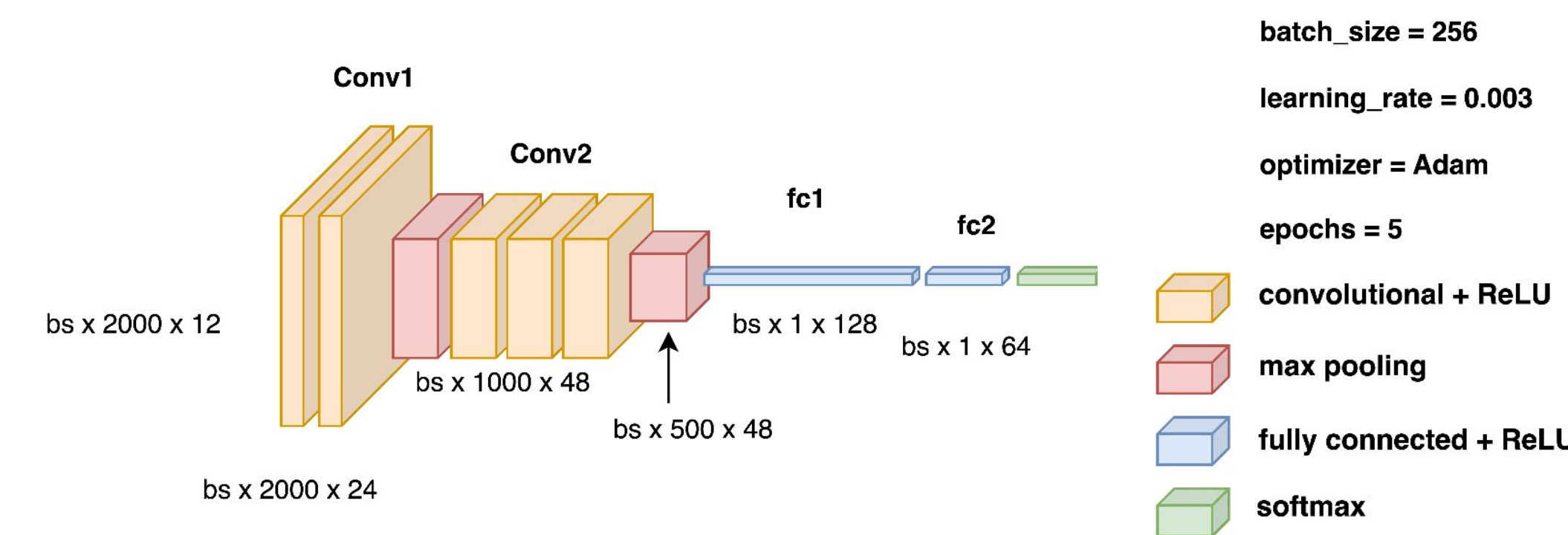


Figure 3: CNN-based architecture for highest accuracy model.

Findings

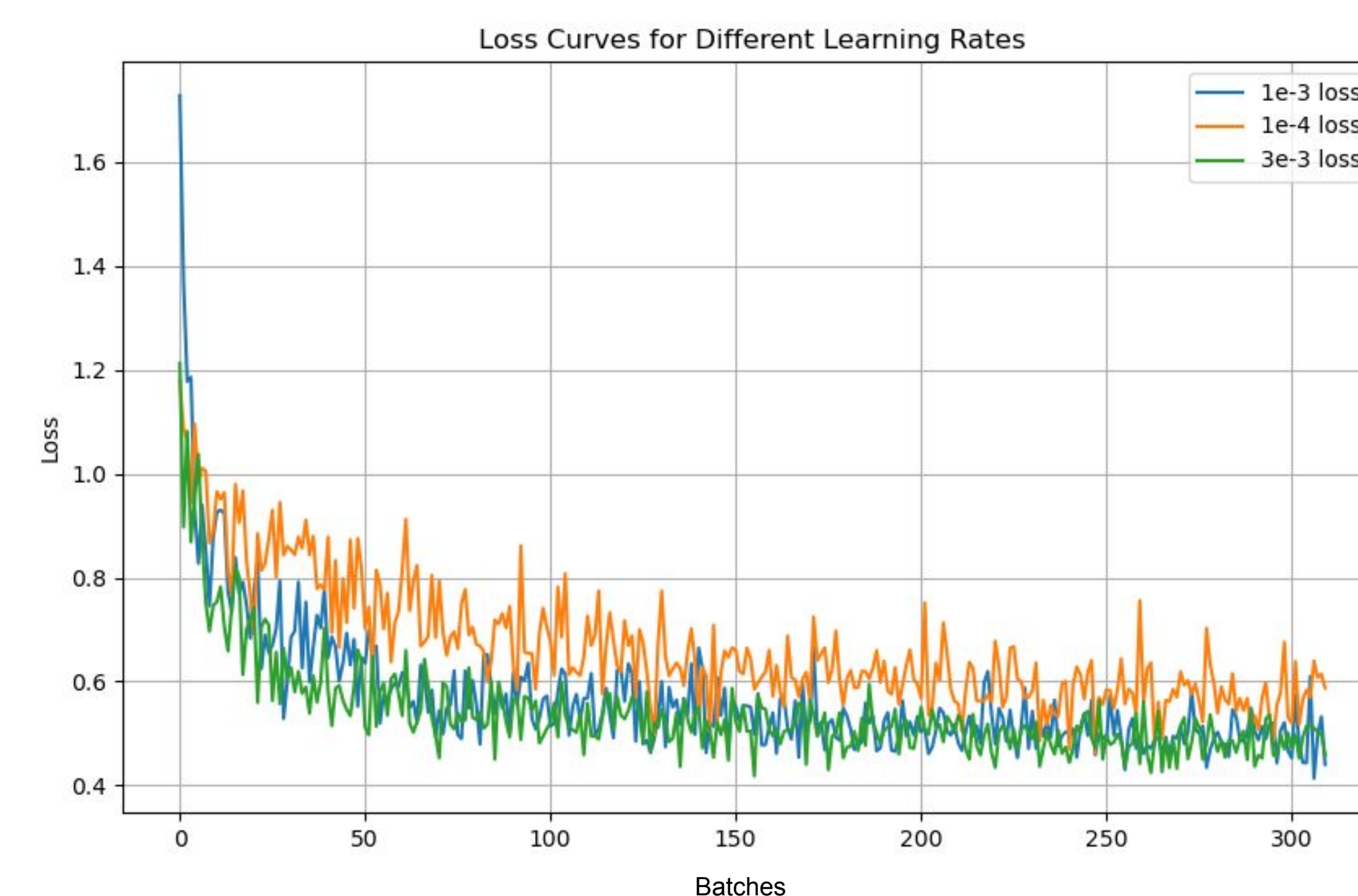


Figure 4. Impact of learning rate on training loss

Learning rate of 3e-3 performed the best for our CNN model

Figure 5. Chagas Detection by CNN Confusion Matrix

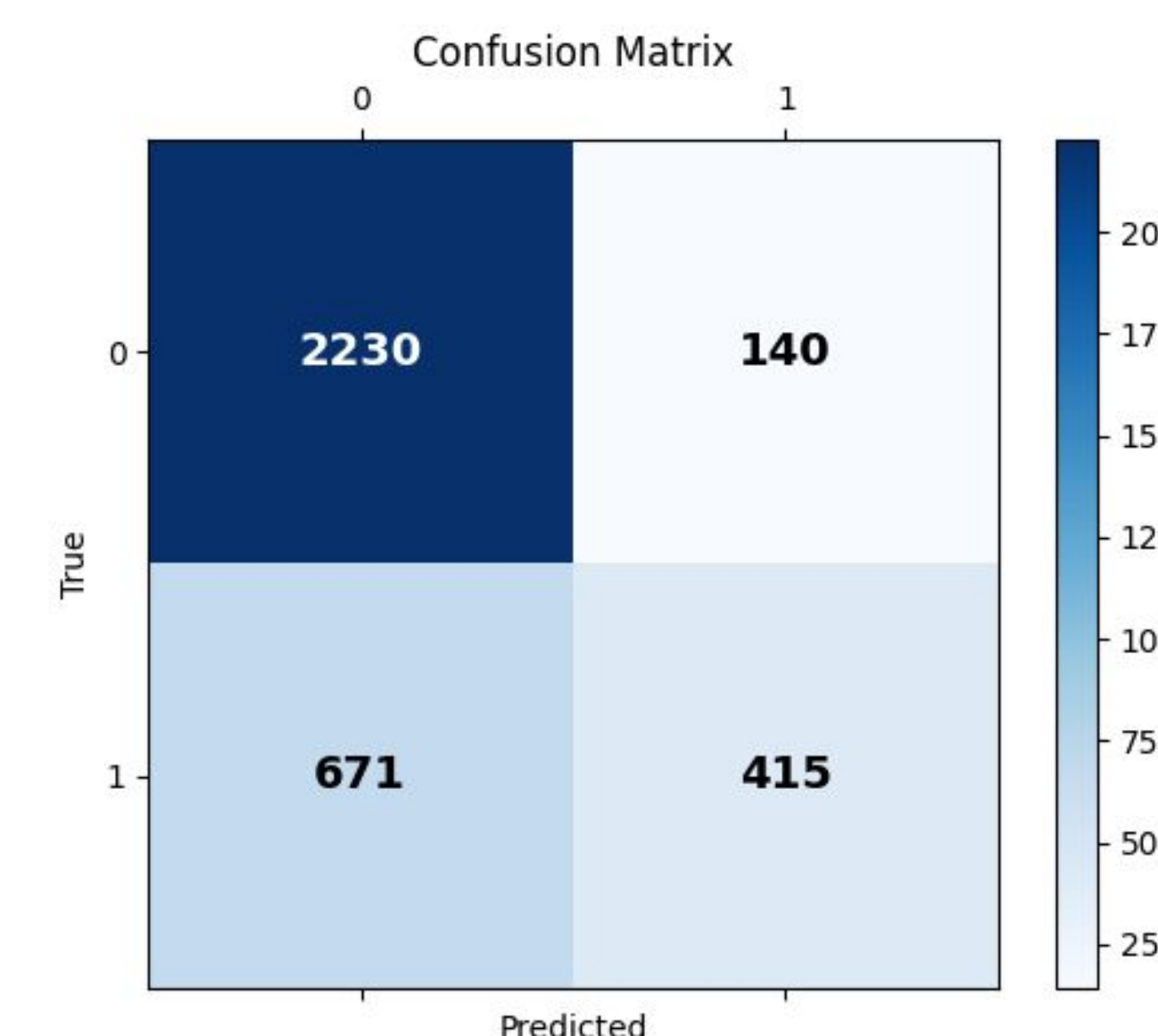


Figure 6. Chagas Detection Performance Breakdown

Overall Accuracy	76.5%
True Positive (Sensitivity)	38.2%
True Negative (Specificity)	94.1%
False Positive	5.9%
False Negative	61.8%

Abstract

Deep learning-based detection of Chagas disease from 12-lead electrocardiograms (ECGs) enables patient identification for confirmatory serological testing. We preprocessed ECG data and compared classification performance for a multilayer perceptron (MLP), convolutional neural network (CNN), and long short-term memory (LSTM) network. Our best model obtained an accuracy of 76.5%, false negative rate of 59.8%, and specificity of 94.1%. We provide a methodology to developing clinical diagnostic tools that could be particularly useful in resource-limited communities.

Discussion

- Best performing models reached accuracies upwards of 78% on unseen testing data.
- Hyperparameter tuning of CNN models revealed best performances at a learning rate to be 3e-3 using batch sizes of 256 (batch size 264 had best performance when compared to other batch sizes).
- Began developing an LSTM model but were limited by time and computational resources. Were able to run one epoch and achieved an training accuracy of 68%. MLP had accuracy of 70%.
- Faced significant challenges stemming from the data provided. Originally obtained accuracies greater than 98% before noticing the true to false ratio in the provided data was 2:98. As a result, spent more time than expected downloading data. Spent hours addressing memory issues and hacking a way to download all the data onto a team member's laptop.
- Currently developing an LSTM and MLP hybrid model integrating ECG waveforms with the patient metadata. Future steps include additional preprocessing of waveform data (such as outlier removal),
- Implications: Efficient (particularly, highly sensitive) detection of Chagas cardiomyopathies holds potential to reduce enduring health infrastructural disparities between rural and urban populations — enabling quick diagnosis, appropriate investment of resources and portable diagnostic centers.

Acknowledgements

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References

